

Cloud Security Risk Assessment Report

Microsoft Azure Governance, Risk & Compliance (GRC) Assessment

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Assessment Type: Cloud Security Risk Assessment & Governance Review

Environment: Microsoft Azure

1. Executive Summary

This report presents the results of a Governance, Risk, and Compliance (GRC) assessment conducted against a Microsoft Azure-hosted Windows Server virtual machine environment.

The objective of this assessment was to evaluate the security posture of a newly deployed Azure infrastructure by identifying assets, vulnerabilities, threats, and security control gaps. The assessment focused on cloud governance, access control, monitoring, network security, and compliance alignment rather than penetration testing or exploitation activities.

The review identified several medium-to-high risk findings including:

- Publicly exposed Remote Desktop Protocol (RDP)
- Absence of Multi-Factor Authentication (MFA)
- Single privileged account dependency
- Limited monitoring and alerting configuration
- Default operating system deployment configuration

No evidence of compromise was observed during the assessment; however, the identified weaknesses represent common cloud security misconfigurations frequently targeted by threat actors.

Overall Risk Rating: Medium-High

2. Assessment Scope

In Scope

- Azure Subscription
- Azure Resource Group
- Azure Virtual Machine
- Virtual Network
- Network Security Group
- Public IP Configuration
- Windows Server Operating System
- Local Administrative Accounts
- Monitoring & Logging Configuration

Out of Scope

- Azure Active Directory Tenant Administration
- Production Applications
- Databases
- Third-Party Integrations
- Penetration Testing Activities

3. Environment Overview

Environment Details

Component	Description
Cloud Provider	Microsoft Azure
Resource Group	GRC-Lab
Virtual Machine	GRC-WIN-VM01
Operating System	Windows Server 2025 Datacenter
Region	Poland Central
VM Size	Standard B2ats v2
Public IP	Enabled
Remote Access	RDP (TCP 3389)
Monitoring	Azure Monitor
Logging	Windows Security Event Logs

Evidence

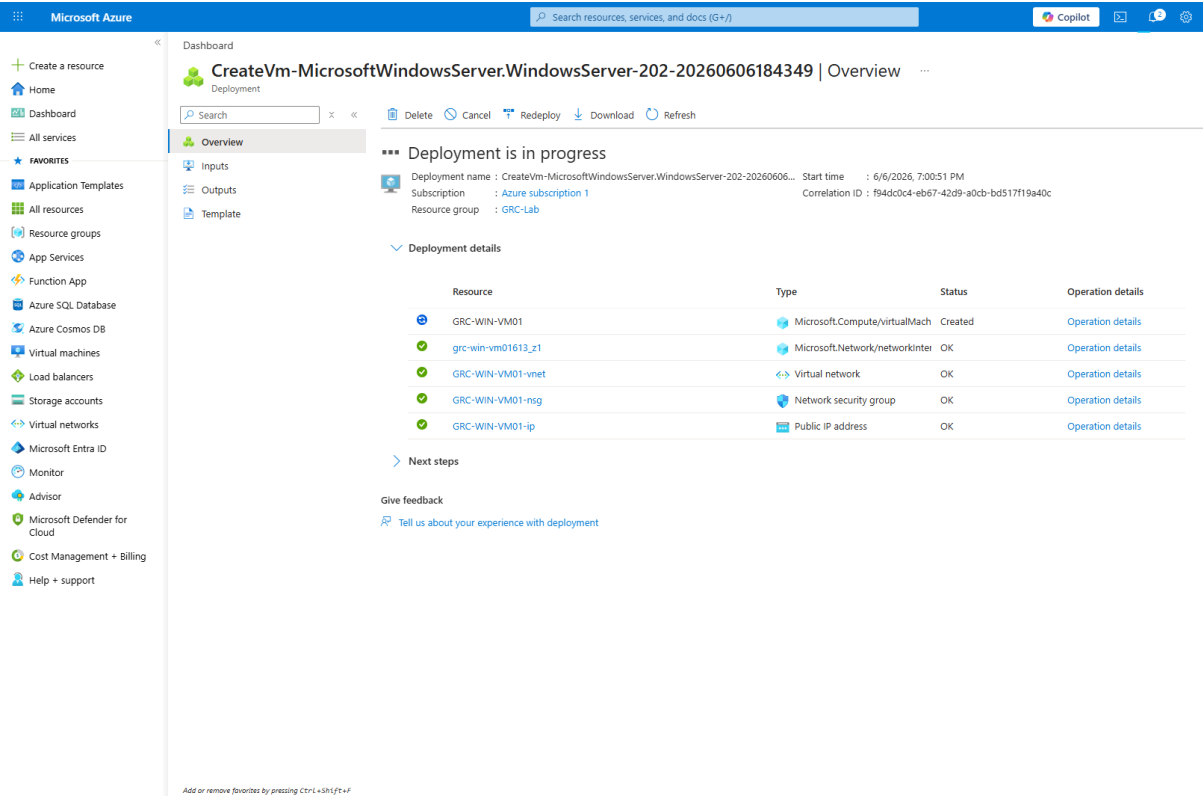


Figure 1. Azure Virtual Machine Creation

Screenshot: VM creation wizard showing deployment of GRC-WIN-VM01 in Azure.

Figure 2. Azure Network Configuration

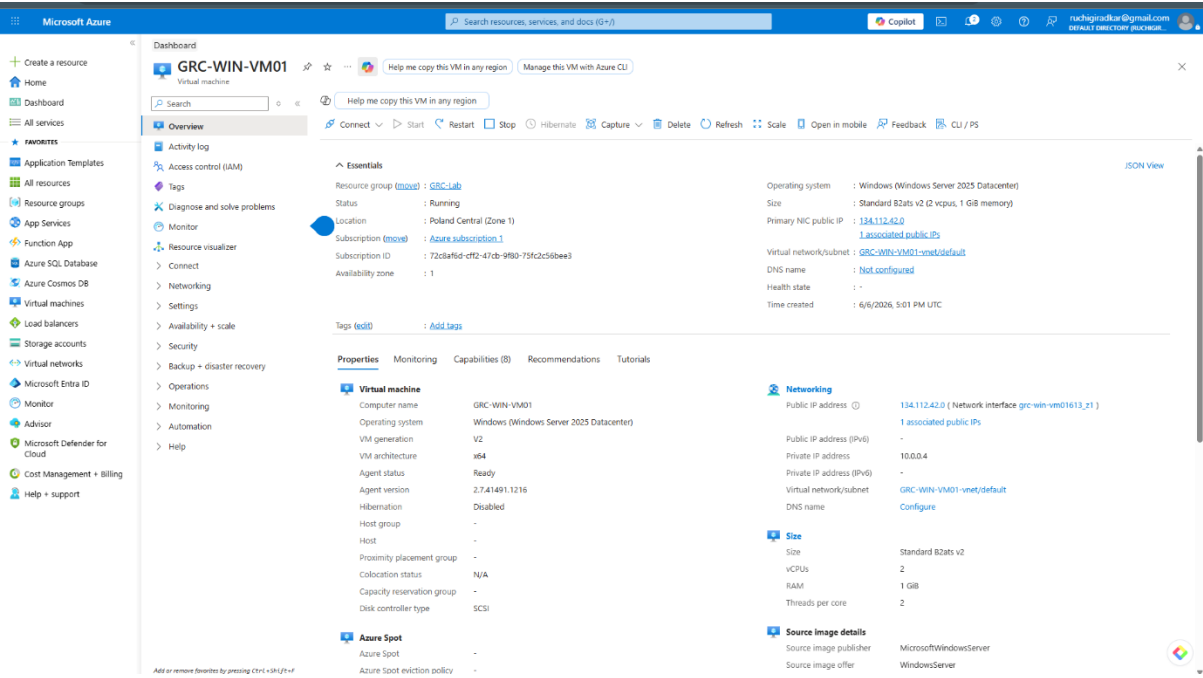
The screenshot displays the 'Create a virtual machine' wizard in the Azure portal, specifically the 'Networking' tab. The left sidebar shows the Azure navigation menu with 'Virtual machines' selected. The main content area is titled 'Create a virtual machine' and includes a search bar and a 'Help me choose the right VM size for my workload' button. Below the title are tabs for 'Basics', 'Disks', 'Networking' (active), 'Management', 'Monitoring', 'Advanced', 'Tags', and 'Review + create'. A descriptive paragraph explains that the user is defining network connectivity by configuring NIC settings. The 'Network interface' section states that a network interface will be created. The configuration fields are as follows:

- Virtual network ***: (new) GRC-WIN-VM01-vnet (with a 'Create new' link)
- Subnet ***: (new) default (10.0.0.0/24)
- Public IP**: (new) GRC-WIN-VM01-ip (with a 'Create new' link and a note: 'Public IP addresses have a nominal charge. [Estimate price](#)')
- NIC network security group**: Basic (selected), with options for None and Advanced.
- Public inbound ports ***: Allow selected ports (selected), with an option for None.
- Select inbound ports ***: RDP (3389)
- Warning box**: 'This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.'
- Delete public IP and NIC when VM is deleted**: Unchecked checkbox.
- Enable accelerated networking**: Unchecked checkbox, with a note: 'The resource provider 'Microsoft.Network' should be registered in order to enable accelerated networking. [Learn more](#)'.

At the bottom, there is a 'Load balancing' section and navigation buttons: '< Previous', 'Next : Management >', and 'Review + create'. A 'Give feedback' link is also present.

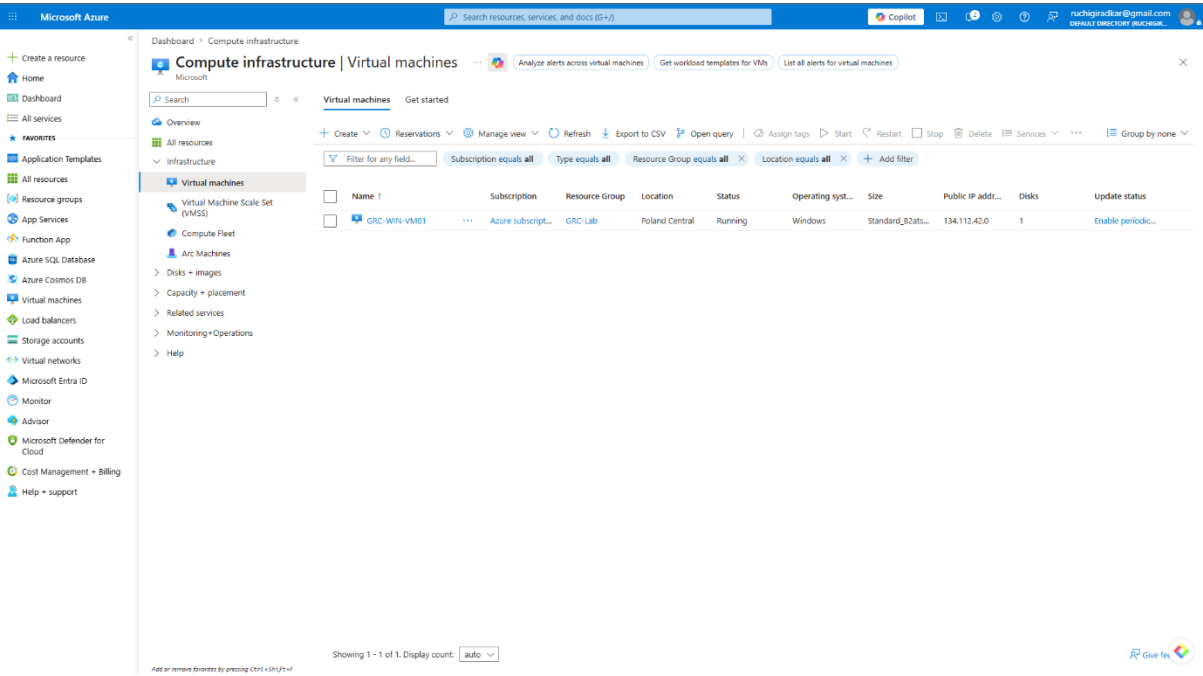
Screenshot showing Virtual Network, Public IP assignment, subnet configuration, and NSG configuration.

Figure 3. Successful Azure Deployment



Screenshot showing successful deployment of Azure VM, NSG, VNet, and Public IP resources.

Figure 4. Azure VM Overview



Screenshot displaying deployed VM properties, operating system, public IP, and resource configuration.

4. Assessment Methodology

The assessment followed a structured Governance, Risk and Compliance framework:

1. Asset Identification
2. Threat Identification
3. Vulnerability Analysis
4. Risk Scoring
5. Security Control Mapping
6. Compliance Review
7. Mitigation Planning

Risk Formula

Risk Score = Likelihood × Impact

Likelihood:

Score Description

- | | |
|---|----------------|
| 1 | Rare |
| 2 | Unlikely |
| 3 | Possible |
| 4 | Likely |
| 5 | Almost Certain |

Impact:

Score Description

- | | |
|---|----------|
| 1 | Minimal |
| 2 | Minor |
| 3 | Moderate |
| 4 | High |
| 5 | Critical |

5. Asset Inventory

Asset Category	Asset	Business Importance
Compute	Azure Virtual Machine	Hosts services and workloads
Identity	Azure Administrator Account	Full administrative control
Network	Public IP Address	External access path
Network	RDP Service	Remote administration
Monitoring	Azure Monitor	Visibility and alerting
Logging	Windows Event Logs	Investigation and auditing
Cloud Management	Azure Subscription	Resource governance

Evidence

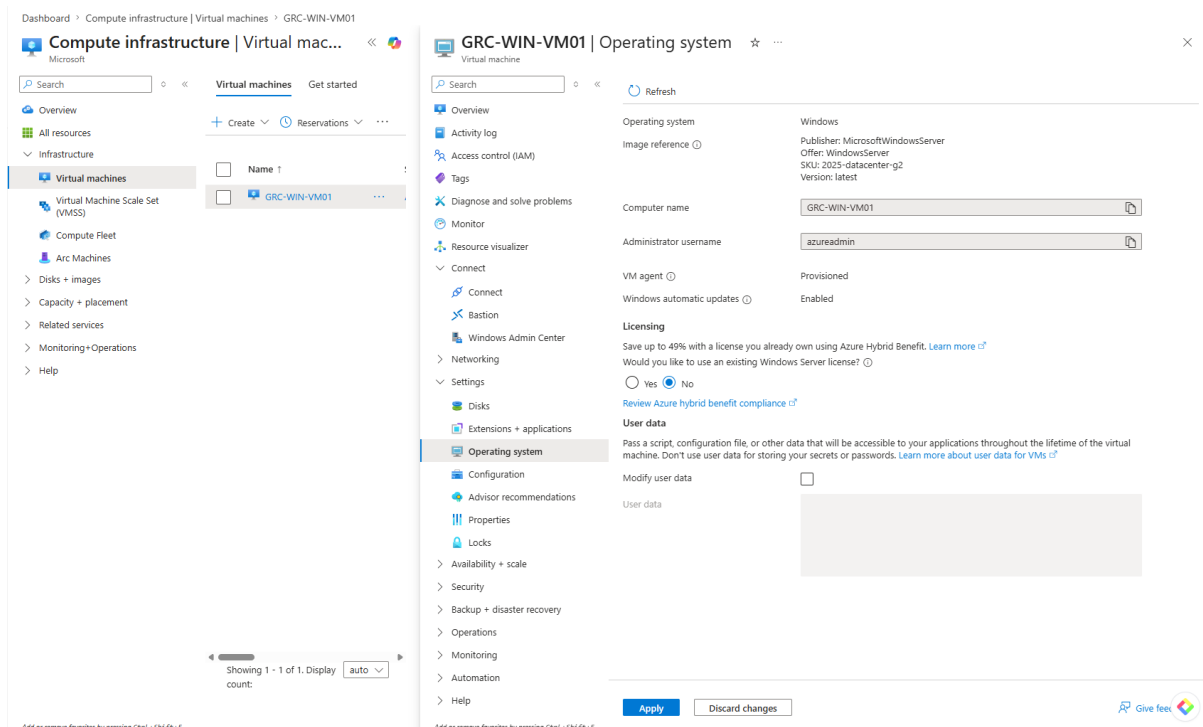


Figure 5. Remote Desktop Connectivity Configuration

Screenshot showing RDP access configuration and public connectivity to the VM.

6. Threat Assessment

Identified Threat Scenarios

Asset	Threat
Public RDP	Brute-force attacks
Admin Account	Credential compromise
Azure Subscription	Unauthorized changes
Operating System	Exploitation of vulnerabilities
Security Logs	Undetected malicious activity

Potential threat actors include:

- Internet-based attackers
- Automated scanning bots
- Credential theft groups
- Insider threats
- Opportunistic threat actors

7. Vulnerability Assessment

Finding 1 – Public RDP Exposure

Observation

The Network Security Group permits inbound TCP/3389 traffic from any source.

Risk

Publicly exposed RDP services are among the most commonly targeted services in cloud environments and are frequently subjected to brute-force and credential-stuffing attacks.

Severity

High

Evidence

The screenshot shows the Microsoft Azure portal interface for creating a virtual machine. The left sidebar contains navigation links such as 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area is titled 'Create a virtual machine' and includes a search bar and a 'Help me choose the right VM size for my workload' button. The 'Review + create' tab is selected, showing the price of 0.0200 USD/hr and the terms of service. A warning message is displayed: 'You have set RDP port(s) open to the internet. This is only recommended for testing. If you want to change this setting, go back to Basics tab.' The 'Basics' tab shows configuration details like Subscription, Resource group, Virtual machine name, Region, Availability options, Zone options, Availability zone, Security type, Enable secure boot, and Enable vTPM.

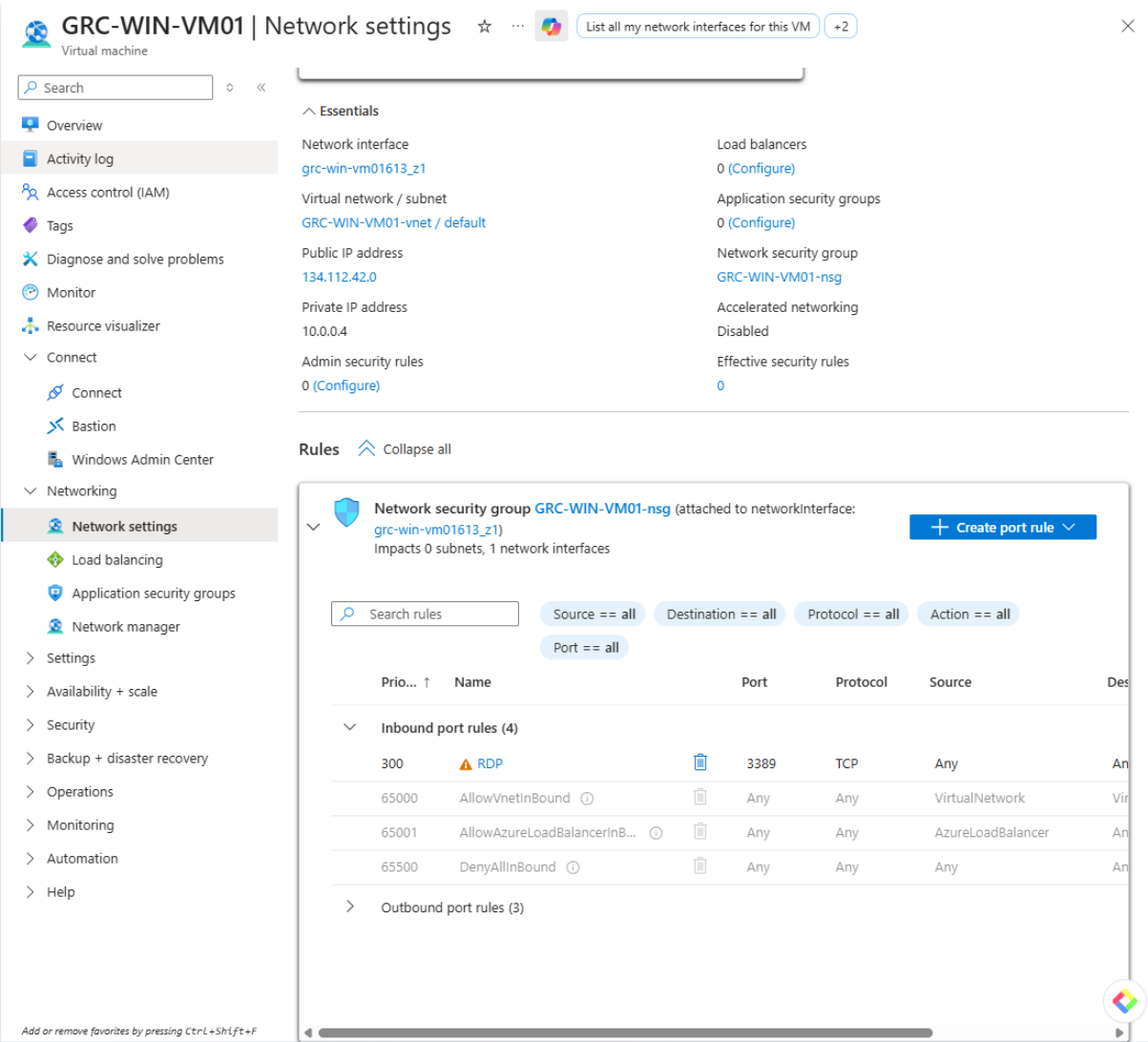
Basics	
Subscription	Azure subscription 1
Resource group	GRC-Lab
Virtual machine name	GRC-WIN-VM01
Region	Poland Central
Availability options	Availability zone
Zone options	Self-selected zone
Availability zone	1
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes

Figure 6. Azure Warning During Deployment

Azure warning message indicating that RDP is exposed to the internet.

Figure 7. NSG Rule Allowing RDP Access

Network Security Group rule permitting inbound RDP access from any source.



Finding 2 – Lack of Multi-Factor Authentication

Observation

Administrative access relies on a username and password without evidence of MFA enforcement.

Risk

Compromised credentials could provide unauthorized privileged access.

Severity

High

Finding 3 – Default Operating System Configuration

Observation

Windows Server was deployed using baseline default settings.

Risk

Default configurations increase exposure to known attack techniques.

Severity

Medium

Evidence

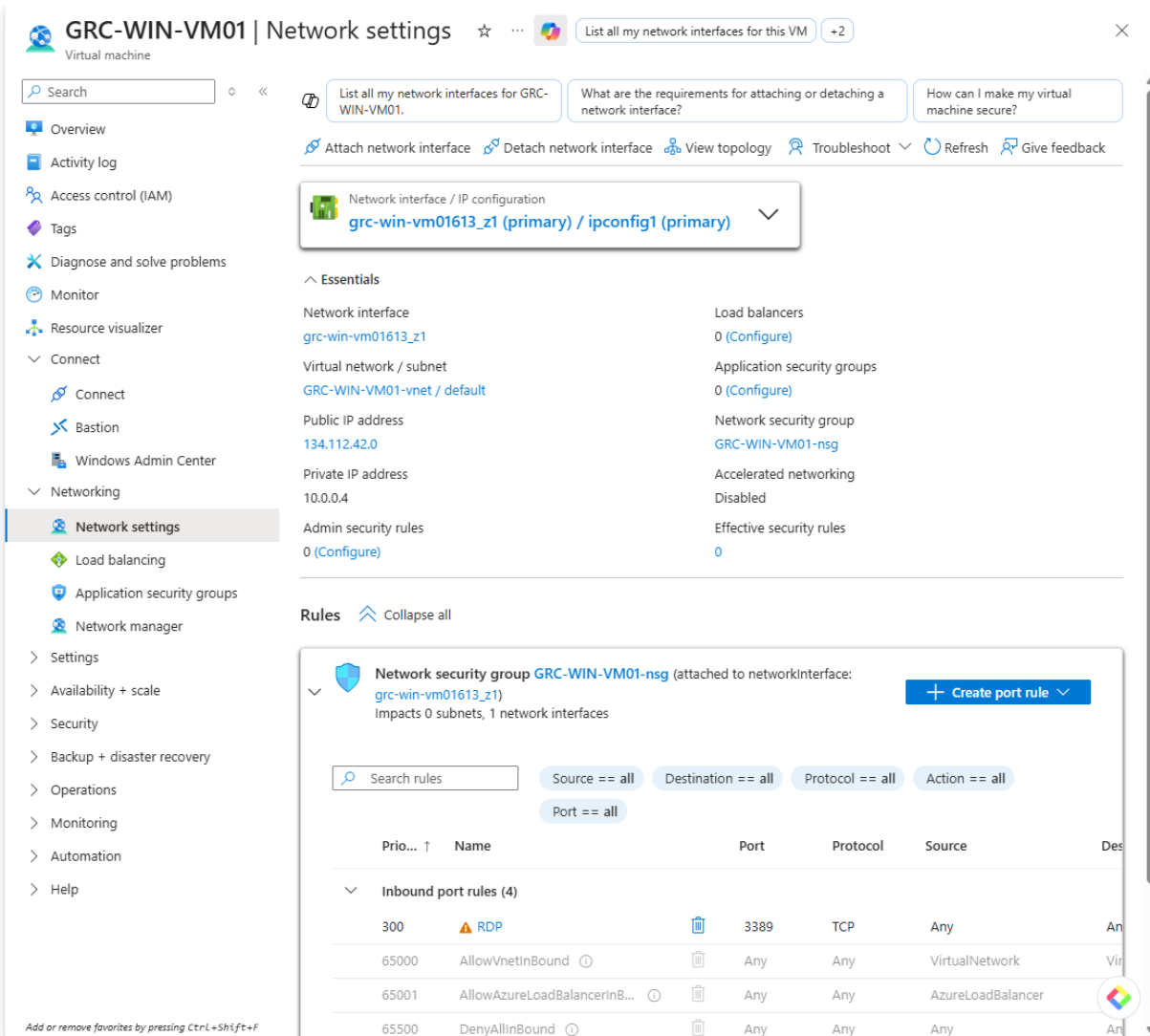


Figure 8. Operating System Configuration

Windows Server configuration displaying administrative account and deployment settings.

Finding 4 – Limited Monitoring and Alerting

Observation

Azure Monitor alerting was not configured.

Risk

Delayed detection and response to security incidents.

Severity

Medium

Evidence

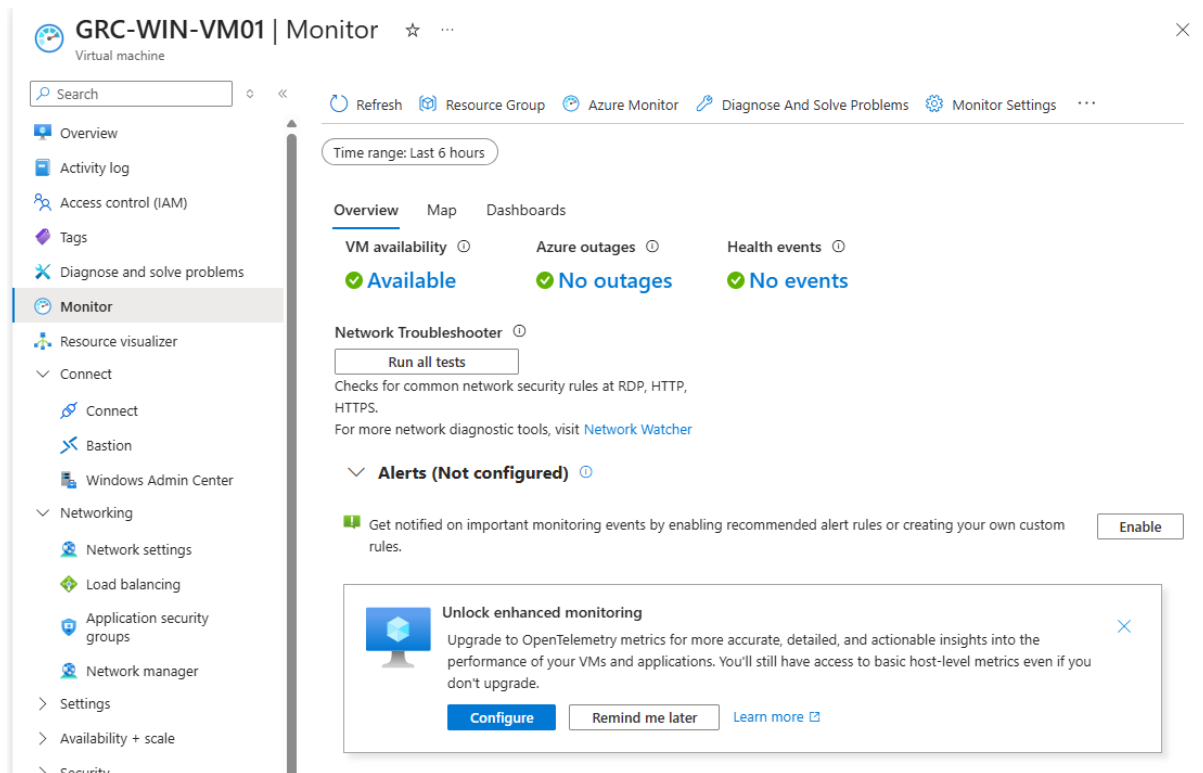
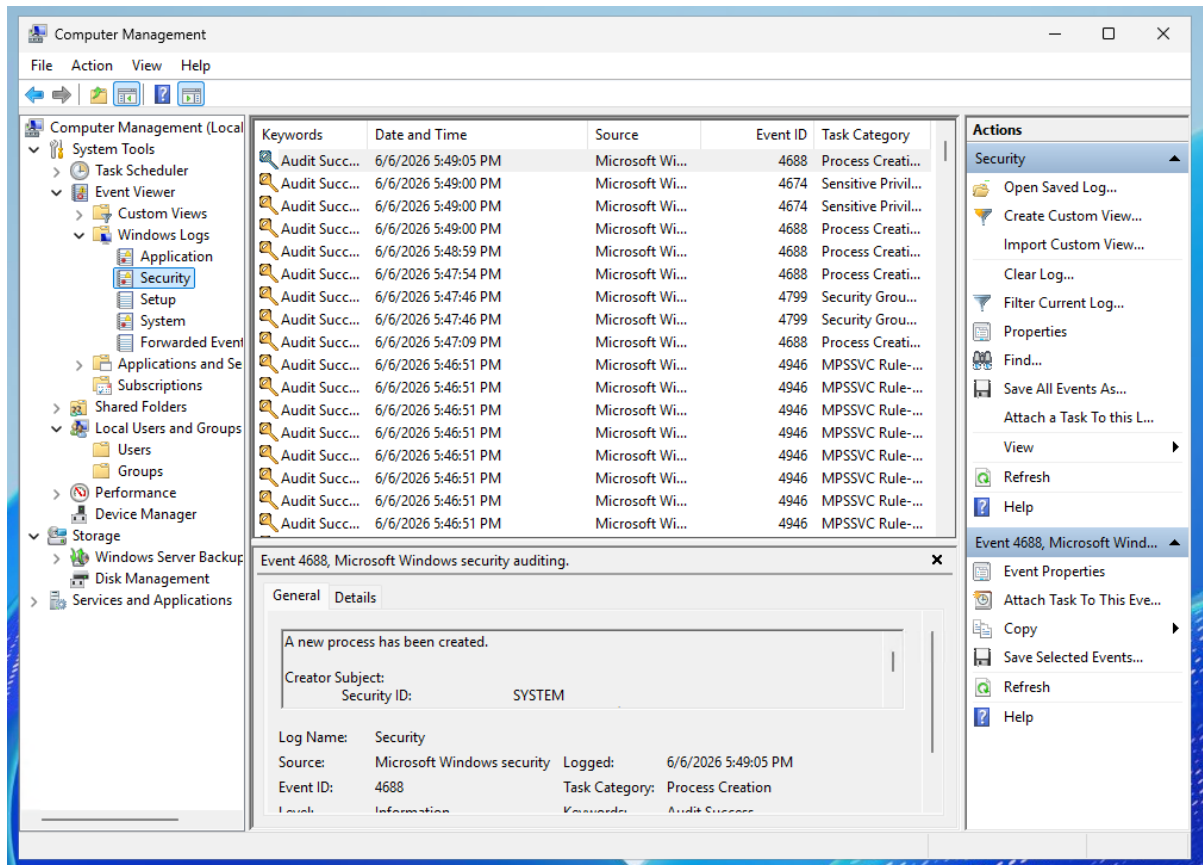


Figure 9. Azure Monitor Configuration

Azure Monitor dashboard showing monitoring status and absence of configured alerts.

Figure 10. Windows Security Event Logs

Windows Security Event Viewer displaying audit events generated by the server.



8. Risk Register

Asset	Threat	Likelihood	Impact	Score	Rating
Public RDP	Brute-force attack	4	4	16	Critical
Admin Account	Credential theft	3	5	15	Critical
Operating System	Vulnerability exploitation	3	4	12	High
Azure Subscription	Unauthorized changes	2	5	10	Medium
Security Monitoring	Delayed detection	3	3	9	Medium

9. Security Control Mapping

Risk	Control	Type	Framework
RDP Exposure	Restrict NSG rules	Preventive	CIS Control 4
Credential Theft	MFA Enforcement	Preventive	NIST CSF PR.AC
Vulnerability Exploitation	Patch Management	Corrective	CIS Control 7
Lack of Monitoring	Centralized Logging	Detective	NIST CSF DE.CM
Privilege Abuse	RBAC Implementation	Preventive	CIS Control 5

10. Recommended Mitigation Actions

Immediate Actions

Priority 1

- Restrict RDP access to approved IP ranges
- Implement Azure Bastion
- Enable MFA for all privileged accounts

Priority 2

- Enable Microsoft Defender for Cloud
- Configure Azure Monitor Alerts
- Enable centralized logging

Priority 3

- Implement RBAC
- Establish hardening baselines
- Conduct periodic security reviews

11. Business Impact

Failure to remediate identified risks may result in:

- Unauthorized access
- Service disruption
- Compliance issues
- Increased recovery costs
- Reputational damage
- Data confidentiality risks

12. Evidence Summary

Evidence ID	Description
E-01	Azure VM Creation
E-02	Network Configuration
E-03	Successful Deployment
E-04	VM Overview
E-05	RDP Connectivity
E-06	RDP Exposure Warning
E-07	NSG Rule Review
E-08	OS Configuration
E-09	Azure Monitor Review
E-10	Windows Security Logs

13. Conclusion

This assessment demonstrates the application of Governance, Risk, and Compliance principles to a Microsoft Azure cloud environment. The review identified common cloud security risks, including exposed remote access services, identity management weaknesses, and monitoring gaps.

The assessment successfully translated technical findings into measurable business risks and aligned recommendations with recognized frameworks such as:

- NIST Cybersecurity Framework (CSF)
- CIS Critical Security Controls
- ISO/IEC 27001

The resulting recommendations provide a practical roadmap for improving security posture while demonstrating cloud security governance, risk assessment, and security control evaluation capabilities.